# Basic Electronics: Ohm's Law and Resistors

Quick notes (sample)

## 1. Ohm's law and power

```
V = I x R
P = V x I = I^2 x R = V^2 / R
```

## 2. Combining resistors

| Connection | Equivalent resistance |
|---|---|
| Series | R = R1 + R2 + R3 |
| Parallel | 1/R = 1/R1 + 1/R2 + 1/R3 |

## 3. Worked examples

Example 1: 6 V is applied across a 2 kilo-ohm resistor.

```
I = V / R = 6 / 2000 = 3 mA
P = V x I = 6 x 0.003 = 18 mW
```

Example 2: 100 ohm and 220 ohm in series, and two 100 ohm resistors in parallel.

```
series:   100 + 220 = 320 ohm
parallel: 1/R = 1/100 + 1/100 = 2/100, so R = 50 ohm
```

Example 3: choosing an LED resistor from a 5 V supply. LED drop 2 V, current 20 mA.

```
R = (5 - 2) / 0.020 = 150 ohm
```

Always check the power rating of a resistor: here P = I^2 R = 0.02^2 x 150 = 0.06 W, so a quarter-watt resistor is enough.